Math 344, F18 Name: Answer

Quiz 11

<u>Instructions:</u> **There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.**

Problem 1. (5 points) Find the determinant of $\mathbf{A}$ using the co-factor expansion where

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 3 & 4 & 5 \\ 5 & 4 & 0 & 3 \\ 2 & 0 & 0 & 1 \end{bmatrix}.$$

$$\det A = \begin{vmatrix} 1 & 0 & 0 & 2 \\ 0 & 3 & 4 & 5 \\ 5 & 4 & 0 & 3 \\ 2 & 0 & 0 & 1 \end{vmatrix} = 1 \cdot \begin{vmatrix} 3 & 4 & 5 \\ 4 & 0 & 3 \\ 0 & 0 & 1 \end{vmatrix} - 2 \begin{vmatrix} 0 & 3 & 4 \\ 5 & 4 & 0 \\ 2 & 0 & 0 \end{vmatrix}$$

$$= 1 \cdot 1 \cdot \begin{vmatrix} 3 & 4 \\ 4 & 0 \end{vmatrix} - 2 \cdot 2 \begin{vmatrix} 3 & 4 \\ 4 & 0 \end{vmatrix}$$

$$= (3 \cdot 0 - 4 \cdot 4) - 4(3 \cdot 0 - 4 \cdot 4)$$

$$= -16 + 64$$

$$= 48$$

Problem 2. (5 points) Let $\mathbf{A} = \begin{bmatrix} 0 & 1 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$. Solve the linear system $\mathbf{Ax} = \mathbf{b}$ with $\mathbf{b} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ by **Cramer's Rule**.

$$\det A = \begin{vmatrix} 0 & 1 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{vmatrix} = (-1)\cdot\begin{vmatrix} 1 & 3 \\ 1 & 1 \end{vmatrix} + 2\cdot\begin{vmatrix} 1 & 3 \\ 0 & 1 \end{vmatrix} = (-1)\cdot(-2)+2\cdot 1 = 4$$

$$B_1 = \begin{bmatrix} 1 & 1 & 3 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \Rightarrow \det B_1 = 1\cdot\begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = -1 \Rightarrow x_1 = \frac{\det B_1}{\det A} = -\frac{1}{4}$$

$$B_2 = \begin{bmatrix} 0 & 1 & 3 \\ 1 & 0 & 1 \\ 2 & 0 & 1 \end{bmatrix} \Rightarrow \det B_2 = (-1)\cdot\begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} = 1 \Rightarrow x_2 = \frac{\det B_2}{\det A} = \frac{1}{4}$$

$$B_3 = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 2 & 1 & 0 \end{bmatrix} \Rightarrow \det B_3 = 1\cdot\begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} = 1 \Rightarrow x_3 = \frac{\det B_3}{\det A} = \frac{1}{4}$$